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TCMS - Operational-VF.CONCEPT

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1 Introduction

This document is a function specification for the control of the TCMS Operational system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

BA	Bus Administrator
CD	Circuit Diagram
CCU	Central Control Unit
VCU	Vehicle Control Unit
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
ETB	Ethernet Train Backbone
ECN	Ethernet Consist Network (ring)
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
IPT	Internet Protocol - Train
MCG	Mobile Communication Gateway
NTP	Network Time Protocol
TCMS	Train Control and Management System
TWCS	Train to Wayside Communication System
MVB	Multifunctional Vehicle Bus
WTB	Wired Train Bus

2 System Overview

The task for the TCMS system is to provide a platform for software controlled functionalities of the train including; Controllers (VCU, IDU etc.), Communication between controllers and Interface to external electrical equipment.

The TCMS System identifies the train configuration and make this information available for the software controlled functions in the train.

The TCMS Operational is reading/writing signals from/to external electrical systems and makes that information available for functionality implemented in the VCU.
In addition some base functionality is includes (time synchronization, redundancy, etc).

Note: MCG not included, part of TWCS.

2.1 System Configuration

Failure Modes:

The following failure modes are defined within TCMS electronics. Each individual mode is detected by the unit and reported to the application software. The application software generates individual events for those failures. TCMS redundancy is designed to cope with any single point failure. Hence as long as one event is occurring stand alone, it does not affect the operation of the vehicle.

1. Component failure: Any TCMS electronics unit is defective. All in/outputs are de-energised. Error may be caused by power supply failure, supply connector failure, CPU failure.
2. Failure of specific input step: Applicable to MIO, the actual input gets high impedative, the inputs data is neutralised in software.
3. Failure of specific output step: Applicable to MIO, the actual output gets high impedative to neutralise the effect on the electrical circuits.
4. IDU component failure: failures on sub-components as touch screen, background light or CPU will cause a total IDU failure.
5. Cable and Contactor Socket failures are not classified as TCMS failures, they are rather depending on manufacturing problems. But as they will effect the TCMS performance they are considered.

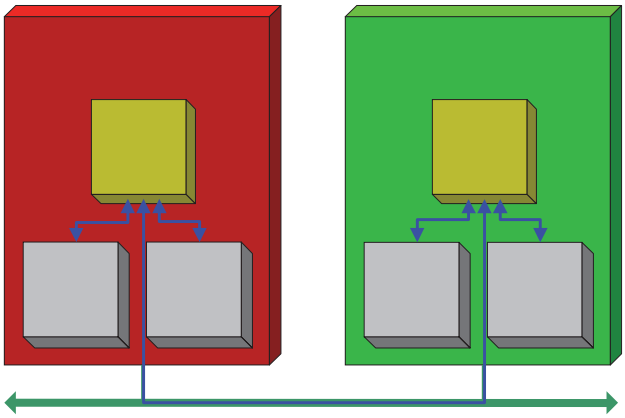
Redundancy Concept:

The following redundancy measures have been taken to improve the system reliability:

1. Redundant MVB: the MVB is of type "dual line EMD" to cope with single point failures in the contactor sockets or in the MVB cabling. In addition two redundant MVB buses are used.
2. Redundant BA running in the CCU: this is a functional redundancy to keep full MVB performance in case of a single BA failure.
3. Redundant I/O: important digital inputs from the train wiring have been duplicated to cope with single cable or TCMS MIO component failures. Duplicated signals are e.g. the input from the cab activation key switch.
4. ECN redundancy: this redundancy is built on a ring structure. The IP ring communication performance will be kept in case a single ring switch or cable failure occurs.
5. ETB redundancy: the ETB is connected using two separate communication cables between each train switch / train repeater.
6. Redundant WTB: the WTB is connected using two separate communication cables between the TCN GWs.
7. Redundant IDU: the primary IDU is the interface to the driver, in case this fails it is possible to activate (by replacing) the back-up IDU.

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8. Redundant CCU:
   The CCU-O is redundant (warm standby), if one CCU-O as a component, or a function within the CCU-O fails the redundant unit will take over (leader - follower principle as shown on the picture below).
   The CCU-D is redundant ensuring that the diagnostic database is still accessible (from redundant device) if one fails.
   The CCU-S is redundant (warm standby), if one CCU-S enters fail safe state the redundant CCU-S take over the function (leader - follower principle as shown on the picture below).
9. Redundant TCN GW: The TCN GW is redundant (warm standby), if one TCN GW as a component, or a function within the TCN GW fails the redundant unit will take over (leader - follower prinicple as shown on the picture below).



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3 Functional requirements

3.1 Architecture

3.1.1 Function Design

The TCMS architecture drawing defines to structure of the TCMS used. In addition to that software architecture elements are used to split up different logical parts in the application software.

3.1.2 Requirements

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TCMS IPT units

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CSS based devices
Units with several cyclic tasks

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3.2.2 Redundancy

3.2.2.1 Function Design

To support different degraded scenarios redundancy is implemented to ensure that impact of single faults are mitigated.

3.2.2.2 Requirements

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3.2.3.2 Requirements

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- the MIO-S shall not be reset automatically, the important functions use (availability-) redundant
- the CCU-S and GW-S shall not be reset automatically, leader-follower redundancy shall cover the single errors for availability.

3.4.2 Normal condition

In normal operation the HMI is working properly and TCMS is not commanding any power cycle. The power supply is connected through a relay normally closed contact.
In case an inconsistency is detected TCMS energizes the relays disconnecting HMI power supply.

3.4.3 Failure cases

1) Command relay is kept energized (IDU not supplied).
2) Command relay faulty (power cycle not possible).

3.4.4 Failure detection

1) Detected by the Driver during operation.
2) Detected by the Driver during operation.

3.4.5 Failure handling

1) Command relay shall be replaced during maintenance.
2) Replace IDU during operation. Command relay to be replaced during maintenance.

3.4.6 Requirements

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

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